

MLFF MVQUAL independent prefix-qualification specification

Status: current normative independent hard-qualification contract

Architecture: revision 105

1. Authority

MVQUAL independently recomputes the hard multi-view coverage and obligation predicates for repaired target-subset prefixes. Its purpose is to verify scientific hard requirements without trusting MVSEL2/REPAIR2 internal counters.

MVQUAL is not a legacy-versus-current comparator and is not a target-size selector. TargetSizeStudyPolicy consumes its current prefix pass/fail evidence.

2. Inputs

A qualification request binds:

- one canonical DATA5 gradient-training domain;
- current DATA7 TargetSubsetInputBundle identity;
- current FEAS1/MVIDX1 primitive identities;
- one current REPAIR2 repaired master-order identity;
- requested prefix size N ;
- exact hard-coverage/obligation policy identity;
- current MVQUAL schema/policy identity.

The selected prefix is exactly the first N entries of the repaired master order. MVQUAL SHALL reject a caller-supplied membership list that does not authenticate as that prefix.

3. Independent recomputation

MVQUAL may reuse authenticated primitive sparse neighborhood/obligation inputs from MVIDX1. It SHALL recompute the qualification predicates independently of selector/repair mutable counters.

For each required family and obligation class, it records the current policy-defined evidence, including as applicable:

- weighted family coverage;
- hard deficit $D_m(N)$ and worst-view deficit $D_{\max}(N)$;
- uncovered witness count and weighted mass;
- required extent/stratum/obligation predicates;
- unique-support and redundancy diagnostics;
- correlation/provenance/run/condition diversity diagnostics;
- exact pass/fail reasons.

Secondary diagnostics cannot compensate for failure of a mandatory hard family or obligation.

4. Hard pass

A prefix passes only when every hard predicate defined by the bound current policy passes. No historical selector/ladder result is part of this comparison.

Locked-test, held-out CV, and calibration evidence cannot tune the qualification policy or satisfy a hard predicate.

5. Monotonicity invariant

Required target-size rungs are nested prefixes of one repaired master order. Under the current positive hard-coverage/obligation predicates, adding candidates cannot remove already covered witness mass or unsatisfy an already satisfied positive obligation.

Therefore, across increasing materializable nominal sizes, the hard-pass sequence SHALL be monotone: once a prefix passes, all larger prefixes under the identical scientific input/policy identity must also pass.

A pass/fail/pass or pass/fail suffix is an invariant violation. MVQUAL SHALL report the inconsistency and target-size preparation SHALL fail closed.

6. Result record

MultiViewQualificationRecord (or the current implementation-equivalent schema) binds:

- domain and prefix size;
- repaired-order identity;
- DATA7/FEAS1/MVIDX1 identities;
- hard policy and MVQUAL policy identities;
- independently recomputed per-family/obligation evidence;
- deterministic pass/fail result and reason codes;
- current schema/version identity.

Execution worker count, queue completion order, cache location, and equivalent runtime choices do not enter the scientific result identity.

7. Parallel execution

Qualification of different domains or prefix sizes may execute concurrently under bounded resource admission. Completion order is non-authoritative; persisted result ordering is canonical.

Parallelism cannot change the primitive sparse relation, FP64 predicate arithmetic, hard thresholds, or pass/fail result.

8. Relationship to target-size study

The target-size study defines:

```
qualified_sizes = {  
  N in materializable_sizes :  
    MVQUAL(d,N).hard_pass for every required training domain d  
}
```

MVQUAL supplies only the independent hard evidence for that definition. It does not train models, rank qualified sizes by learning performance, inspect held-out folds, or choose N_{selected} .

9. Failure conditions

MVQUAL fails closed when:

- primitive MVIDX/policy identity does not match the repaired order;
- requested membership is not the authenticated repaired prefix;
- selector/repair counters are supplied as a substitute for independent recomputation;
- a required hard predicate cannot be evaluated exactly;
- forbidden held-out/calibration/locked evidence enters the qualification policy;
- nested-prefix qualification violates monotonicity;
- persisted qualification evidence is stale, corrupt, or incompatible.

Unsupported historical qualification/comparison schemas are non-current evidence and do not define current acceptance behavior.